Hartree-Fock Theory applied to Jellium

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1 Introduction

We present a summary of several variations of the Hartree-Fock (HF) method applied to jellium. After providing some background theory to motivate our work, we derive the dispersion relation $\epsilon_k(k)$ for jellium and discuss its merits and weaknesses. In particular, we observe that there is an unphysical divergence in the slope of the energy at the Fermi wavevector k_F. We discuss the consensus on this issue and present the hyper-Hartree-Fock (HHF) method which deals with the ground state and the excited states in an average way. We show that the location of the divergent slope in the energy is found at a new, fictitious wavevector k_R which is related to the true Fermi wavevector via a parameter γ.

By varying γ we show that the divergence in the slope of ϵ can be moved along the dispersion relation arbitrarily far from the Fermi wavevector, meaning that, in effect, we can relegate this inconsistency in the HF result to being of negligible importance.

2 Background

2.1 Born-Oppenheimer Approximation

At the fundamental level, the properties of matter are governed by Schrödinger's equation. Solving this equation allows us to accurately describe matter from first principles, and is the focus of a large portion of the field of condensed matter physics. Schrödinger's equation for a general system of nuclei and electrons is

$$\hat{H}_{mol}\ \Psi(\mathbf{r}_1, ..., \mathbf{r}_N; \mathbf{R}_1, ..., \mathbf{R}_N) = E\ \Psi(\mathbf{r}_1, ..., \mathbf{r}_N; \mathbf{R}_1, ..., \mathbf{R}_N). \tag{1}$$

The sets $\{\mathbf{r}_i\}$ and $\{\mathbf{R}_\alpha\}$ represent the positions of electrons and nuclei respectively [1].

Our Hamiltonian is written as follows

$$\begin{aligned} \hat{H}_{mol} &= \hat{H}_e\ + \hat{H}_n \\ &= \hat{T}_e\ + \hat{V}_{ee}\ + \hat{V}_{en}\ + \hat{T}_n\ + \hat{V}_{nn}, \end{aligned} \tag{2}$$

where we use subscripts e and n to denote electrons and nuclei. $\hat{T}$ and $\hat{V}$ represent kinetic and potential energies respectively.

Unfortunately, this equation is impossible to solve either analytically or numerically for all but the simplest systems.

We can simplify (1) by employing the Born-Oppenheimer (BO) approximation [2]. Here we assume that we can decompose the wavefunction as the product

$$\Psi(\mathbf{r}_1, ..., \mathbf{r}_N; \mathbf{R}_1, ..., \mathbf{R}_N) = \psi_{R_1,...,R_N}(\mathbf{r}_1, ..., \mathbf{r}_N) X(\mathbf{R}_1, ..., \mathbf{R}_N). \tag{3}$$

Note that here the $\psi_{R_1,...,R_N}$ depend parametrically on the nuclear positions, and hence this is not an ordinary multiplication of two independent wavefunctions.

The eigenvalue equation (1) can now be separated into two problems, nuclear and electronic. We will only focus on the electronic problem, as it poses the greatest mathematical challenge. Our attention then shifts to solving

$$\hat{H}_e\ \psi_{R_1,...,R_N}(\mathbf{r}_1, ..., \mathbf{r}_N) = \mathcal{E}_k(\mathbf{R}_1, ..., \mathbf{R}_N)\ \psi_{R_1,...,R_N}(\mathbf{r}_1, ..., \mathbf{r}_N). \tag{4}$$

Now our Hamiltonian is the sum of terms as follows

$$\begin{aligned} \hat{H}_e &= \hat{T}_e\ + \hat{V}_{ee}\ + \hat{V}_{en}, \\ &= -\frac{1}{2}\sum_i {\nabla_i}^2\ + \frac{1}{2}\sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{i,\alpha} \frac{Z_\alpha}{|\mathbf{r}_i - \mathbf{R}_\alpha|}, \end{aligned} \tag{5}$$

Here, the set $\{Z_\alpha\}$ refers to the nuclear charges.

2.2 Hartree-Fock Approximation

A useful further approximation is the Hartree-Fock (HF) approximation [3, 4]. In the HF regime, we assume that the many-particle electronic wavefunction $\psi(\{\mathbf{r}_i\})$ can be written as an antisymmetric product of single-particle wavefunctions φ_i, this is known as a Slater determinant [5]. Ignoring spin for the moment, our ansatz is

$$\Phi(\{\mathbf{r}_i\}) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \varphi_1(\mathbf{r}_1) & \dots & \varphi_N(\mathbf{r}_1) \\ \vdots & \ddots & \vdots \\ \varphi_1(\mathbf{r}_N) & \dots & \varphi_N(\mathbf{r}_N) \end{vmatrix} \tag{6}$$

One particular advantage of assuming a solution of this form is that Slater determinants are fermionic by construction, meaning they obey antisymmetry and therefore the Pauli exclusion principle. The factor of $1/\sqrt{N!}$ deals with normalisation. By making this approximation, it is important to note that the electronic wavefunction loses its dependence on the nuclear positions $\{R_\alpha\}$.

The so-called Hartree-Fock method proceeds by employing the Rayleigh-Ritz (RR) variational principle to derive expressions for the single-particle wavefunctions φ_i, often called the orbitals. The RR principle states

$$\frac{\langle\Psi|H|\Psi\rangle}{\langle\Psi|\Psi\rangle} \geq E_o. \tag{7}$$

Through the method of Lagrange multipliers we can derive the HF eigenequations (including spin)

$$\left(-\frac{\nabla^2}{2} + v_{ext}(\mathbf{r}) \ + v_H[\rho](\mathbf{r}) \ + \hat{v}_x[\rho^\sigma]\right)\varphi_k^\sigma = \lambda_k^\sigma \varphi_k^\sigma. \tag{8}$$

In (8) and from here, we will write the electron potential in the Coulomb field of the nuclei V_{en} as the external potential $v_{ext}(\mathbf{r})$. We define the terms in (8) as follows

$$v_{ext}(\mathbf{r}) \ \varphi_k(\mathbf{r}) = -\int \frac{d^3\mathbf{r}'\rho_+(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} \ \varphi_k(\mathbf{r}), \tag{9}$$

$$v_H[\rho](\mathbf{r}) \ \varphi_k(\mathbf{r}) = +\int \frac{d^3\mathbf{r}'\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} \ \varphi_k(\mathbf{r}), \tag{10}$$

$$\hat{v}_x[\rho^\sigma] \ \varphi_k(\mathbf{r}) = -\int \frac{d^3\mathbf{r}'\rho^\sigma(\mathbf{r},\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} \ \varphi_k(\mathbf{r}'). \tag{11}$$

3 Hartree-Fock Method for Jellium

Even after making the BO and HF approximations, (8) remains difficult to solve for real systems. 'Jellium' is the name often given to an idealised system where the nuclear charge is smeared out into a uniform 'jelly'. In the jellium model

we can solve the HF equations analytically, following the well-known approach from the literature [6].

In jellium the HF equations can be written as

$$\left(-\frac{\nabla^2}{2} + v_{ext}(\mathbf{r}) \ + v_H[\rho](\mathbf{r}) \ + \hat{v}_x[\rho^\sigma]\right)\varphi_k = \epsilon_k\varphi_k. \tag{12}$$

The advantage of the jellium model is that the external potential $v_{ext}(\mathbf{r})$ and the Hartree potential $v_H[\rho]$ cancel as follows

$$v_{ext}(\mathbf{r}) \ \varphi_k(\mathbf{r}) = -\int \frac{d^3r'\rho_+(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} = -\frac{N}{V}\int \frac{d^3r'}{|\mathbf{r}-\mathbf{r}'|}, \tag{13}$$

since the uniform positive charge has a density of $+N/V$.

$$v_H[\rho](\mathbf{r}) \ \varphi_k(\mathbf{r}) = +\int \frac{d^3r'\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} = +\frac{N}{V}\int \frac{d^3r'}{|\mathbf{r}-\mathbf{r}'|}, \tag{14}$$

here we have used the fact that $\rho(\mathbf{r}') = +N/V$ for plane wave orbitals since if we assume

$$\varphi_k = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}}, \tag{15}$$

where V is the volume of jellium we are considering, then we have

$$\rho(\mathbf{r}') = \sum_{k=1}^{N}|\varphi_k(\mathbf{r}')|^2 = \sum_{k=1}^{N}\frac{e^{-i\mathbf{k}\cdot\mathbf{r}'}e^{i\mathbf{k}\cdot\mathbf{r}'}}{V} = \sum_{k=1}^{N}\frac{1}{V} = \frac{N}{V}. \tag{16}$$

From this we can clearly see

$$(v_{ext} \ + v_H[\rho]) \ \varphi_k = 0,$$

simplifying equation (12) to

$$\left(-\frac{\nabla^2}{2} + \hat{v}_x[\rho^\sigma]\right)\varphi_k = \epsilon_k\varphi_k. \tag{17}$$

As previously mentioned, we use a trial solution of plane wave orbitals since they greatly simplify the mathematics. For example, if we assume (15), we can deal with the Laplacian term immediately. Namely

$$-\frac{\nabla^2}{2}\varphi_k = \frac{k^2}{2}\varphi_k. \tag{18}$$

This leaves us with the computationally expensive Fock exchange term $\hat{v}_x[\rho^\sigma]\varphi_k$ to calculate.

$$\hat{v}_x[\rho^\sigma] \ \varphi_k(\mathbf{r}) = -\int \frac{d^3r'\rho^\sigma(\mathbf{r},\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} \ \varphi_k(\mathbf{r}'). \tag{19}$$

As is widely known, plane waves are eigenfunctions of the Fock exchange operator $\hat{v}_x[\rho^\sigma]$, meaning we can write the eigen-equation

$$\hat{v}_x[\rho^\sigma]\ \varphi_k(\mathbf{r}) = \lambda_k\ \varphi_k(\mathbf{r}). \tag{20}$$

This is a crucial detail since it tells us our choice of plane wave orbitals allows us to solve (17) for the eigen-energies ϵ_k. Following the standard treatment of this equation, we arrive at the well-known expression for λ_k [6],

$$\lambda_k = -\frac{k_F}{\pi}\left(1 + \frac{k_F^2 - k^2}{2kk_F}\ln\left|\frac{k_F + k}{k_F - k}\right|\right), \tag{21}$$

k_F is the Fermi wavevector.

Combining equations (18) and (21) we can write

$$\epsilon_k = \frac{k^2}{2} - \frac{k_F}{\pi}\left(1 + \frac{k_F^2 - k^2}{2kk_F}\ln\left|\frac{k_F + k}{k_F - k}\right|\right), \tag{22}$$

Figure 1 shows a graph of (22) with the energy dispersion of free electrons.

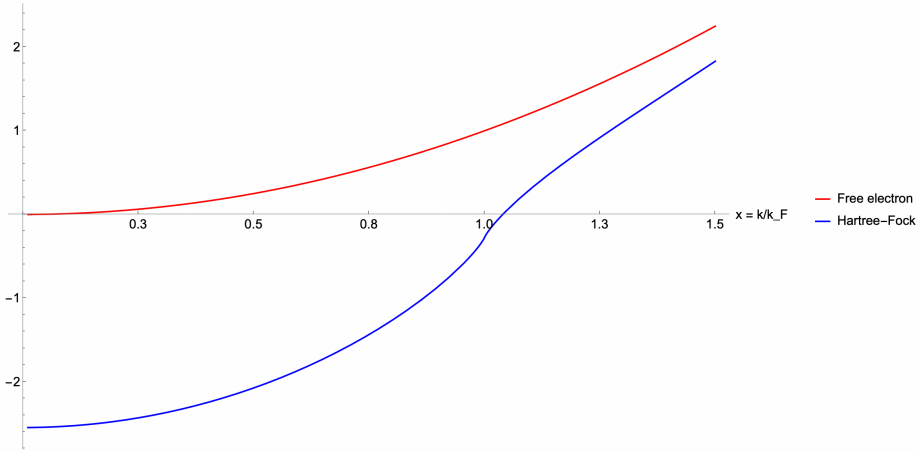


Figure 1: Energy dispersion of the HF approximation (in blue) compared with the free electron (in red).

The key property to note is the divergence in the slope of the HF dispersion relation at the Fermi wavevector $k = k_F$, clearly seen in Figure 2 showing $d\epsilon/dk$ against k.

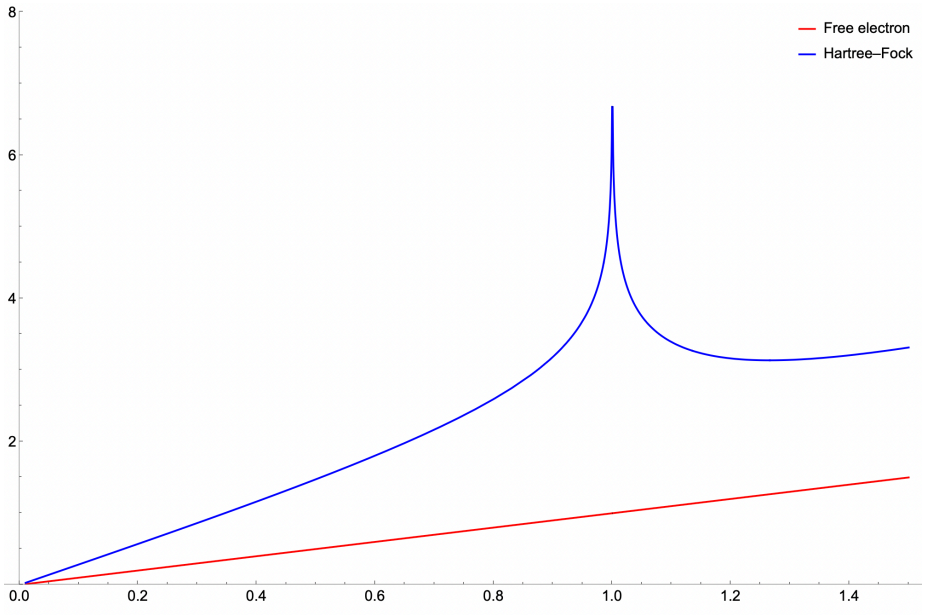


Figure 2: Derivative of the energy dispersion $d\epsilon/dk$ of the HF approximation (in blue) compared with the free electron (in red). The divergence in the slope of $\epsilon(k)$ at the Fermi wavevector k_F is shown.

Recall that the density of states can be written as

$$g(k) \propto \left(\frac{d\epsilon}{dk}\right)^{-1}. \tag{23}$$

With this in mind, it follows that if

$$\frac{d\epsilon}{dk} \to \infty,$$

we have

$$g(k) \to 0. \tag{24}$$

Figure 3 shows the plot of the density of states $g(k)$ against k.

Hence a divergent derivative in the energy implies the existence of an infinitesimal band gap in jellium. It is well-known that jellium is an accurate approximation for many metals, and so this result is unexpected physically.

This failure of the HF method is generally attributed to the lack of incorporating the phenomenon of screening, in which the electronic density adjusts in response to the motion of one electron [7, 8, 9, 10].

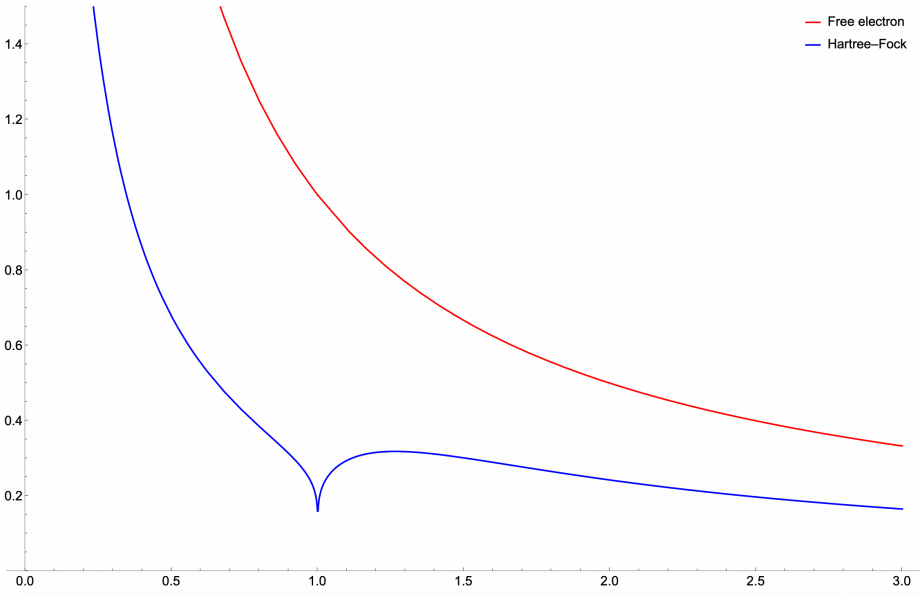


Figure 3: Density of states $g(k)$ against k for the HF approximation in jellium (in blue) compared with the free electron (in red). The zero HF density of states $g(k)$ at the Fermi wavevector is shown.

4 Hyper-Hartree-Fock Method for Jellium

Instead of trying to include screening in an ad-hoc way, we can make an adjustment to the HF method which allows us to resolve our unphysical band gap. This adjustment is commonly referred to as the Hyper-Hartree-Fock theory, in which we consider an N-electron system constructed from a basis set of $R > N$ orbitals. After variationally optimising the average energy, Gidopoulos and Theophilou [11] derived HHF equations similar to (12) as follows

$$\left(-\frac{\nabla^2}{2} + v_{ext}(r) \; + \frac{N-1}{R-1} \sum_{j=1}^{R} \left[J_i(r) - \delta_{s_j,s_i} K_j(r) \right] \right) \varphi_i = \epsilon_i \varphi_i. \tag{25}$$

Solving this HHF equation for jellium follows a similar method to the previous HF method. We begin by trying to combine the Hartree and external potential terms. Consider first the Hartree term

$$J_j(r) \; \varphi_i(r) = \int \frac{d^3r' |\varphi_j(r')|^2}{|r - r'|} \; \varphi_i(r). \tag{26}$$

Define $\gamma = \frac{R-1}{N-1}$ and $\rho_R(r') = (R-1)/V$. We can take the sum inside the integral as follows

$$
\begin{aligned}
v_H(r)\ \varphi_i(r) &= \frac{1}{\gamma}\sum_{j=1,\ j\neq i}^{R} J_j(r)\ \varphi_i(r) \\
&= \frac{1}{\gamma}\sum_{j=1,\ j\neq i}^{R}\int \frac{d^3r'|\varphi_j(r')|^2}{|r-r'|}\ \varphi_i(r) \\
&= \frac{1}{\gamma}\int \frac{d^3r'\sum_{j=1,\ j\neq i}^{R}|\varphi_j(r')|^2}{|r-r'|}\ \varphi_i(r), \\
&= \frac{1}{\gamma}\int \frac{d^3r'\rho_R(r')}{|r-r'|}\varphi_i(r) \\
&= \frac{N-1}{V}\int \frac{d^3r'}{|r-r'|}\ \varphi_i(r). \qquad (27)
\end{aligned}
$$

Now we consider the external potential term.

$$
\begin{aligned}
v_{ext}(r)\ \varphi_i(r) &= -\int \frac{d^3r'\rho_+(r')}{|r-r'|}\ \varphi_i(r), \\
&= -\frac{N}{V}\int \frac{d^3r'}{|r-r'|}\ \varphi_i(r). \qquad (28)
\end{aligned}
$$

Combining these two terms, we arrive at the expression

$$
(v_H(r)\ + v_{ext}(r))\ \varphi_i(r) = -\frac{1}{V}\int \frac{d^3r'}{|r-r'|}\ \varphi_i(r) \qquad (29)
$$

which becomes negligible in the limit as V becomes large. This leaves us with a simplified version of (25).

$$
\left(-\frac{\nabla^2}{2}\ -\frac{1}{\gamma}\sum_{j=1}^{R}\delta_{s_j,s_i}K_j(r)\right)\varphi_i = \epsilon_i\varphi_i \qquad (30)
$$

After summing over j and dealing with the delta can rewrite the second term as

$$
-\frac{1}{\gamma}\sum_{j=1}^{R}\delta_{s_j,s_i}K_j(r)\ \varphi_i = -\frac{1}{\gamma}\int \frac{d^3r'}{|r-r'|}\rho^{\sigma}(r,r')\ \varphi_i(r'). \qquad (31)
$$

From here our HHF problem is analogous to the HF problem previously described but with the addition of a factor of γ in the Fock exchange term. Proceeding with the same derivation as before we arrive at a dispersion relation of

$$
\epsilon_k = \frac{k^2}{2} - \frac{k_R}{\pi\gamma}\left(1 + \frac{k_R^2-k^2}{2kk_R}\ln\left|\frac{k_R+k}{k_R-k}\right|\right). \qquad (32)
$$

It is important to note that k_R here is a fictitious Fermi wavevector of an R electron system. We can write

$$k_R{}^3 = 3\pi^2 \frac{R}{V}. \tag{33}$$

For ease of comparison with our result for the ordinary HF equation (22), it is desirable to express this relation in terms of the real Fermi wavevector k_F. In the thermodynamic limit $N, V \to \infty$ we can write

$$k_R = \gamma^{1/3} k_F. \tag{34}$$

Substituting (34) into (32) we get

$$\epsilon(k) = \frac{k^2}{2} - \frac{k_F}{\pi\gamma^{2/3}} \left(1 + \frac{\gamma^{2/3}k_F^2 - k^2}{2\gamma^{1/3}kk_F} \ln \left| \frac{\gamma^{1/3}k_F + k}{\gamma^{1/3}k_F - k} \right| \right) \tag{35}$$

Figure 5 shows a plot of this result for various values of $\gamma = 1, 2, 5, 10$. Note that in the case $\gamma = 1$ we simply have the ordinary HF result for jellium.

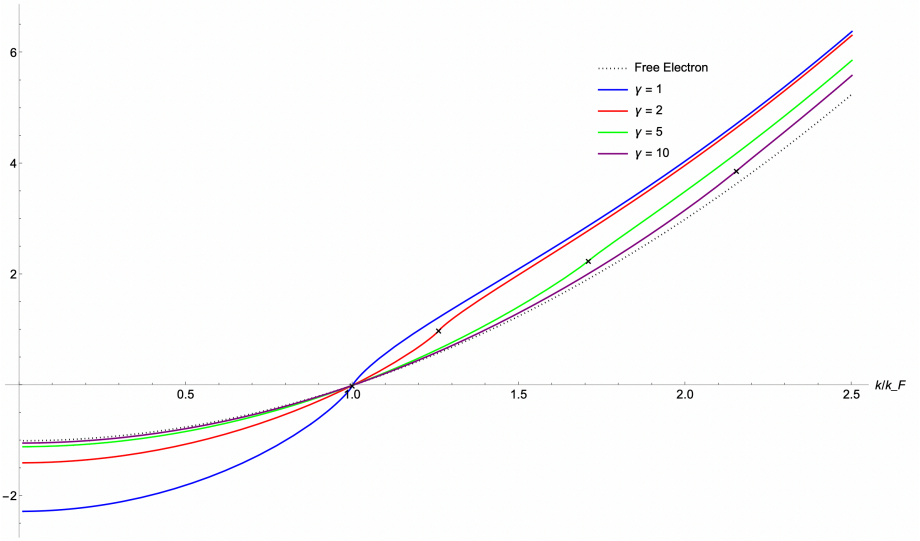


Figure 4: HHF result for jellium. Vertical axis is $[\epsilon(k) - \epsilon(k_F)]/\epsilon_0(k_F)$, where $\epsilon_0(k_F) = {k_F}^2/2$. Crosses mark points with divergent derivatives.

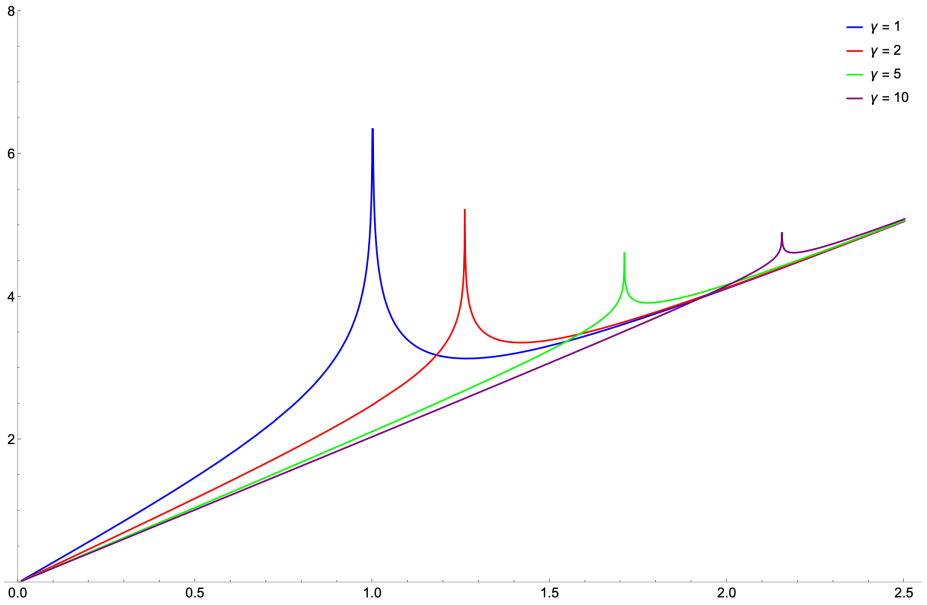


Figure 5: Derivative of the energy dispersion $d\epsilon/dk$ of the HHF approximations for various values of $\gamma = \frac{R-1}{N-1}$. The divergence in the slopes of the various $\epsilon(k)$ at the fictitious Fermi wavevectors $k_R = \gamma^{1/3} k_F$ are shown.

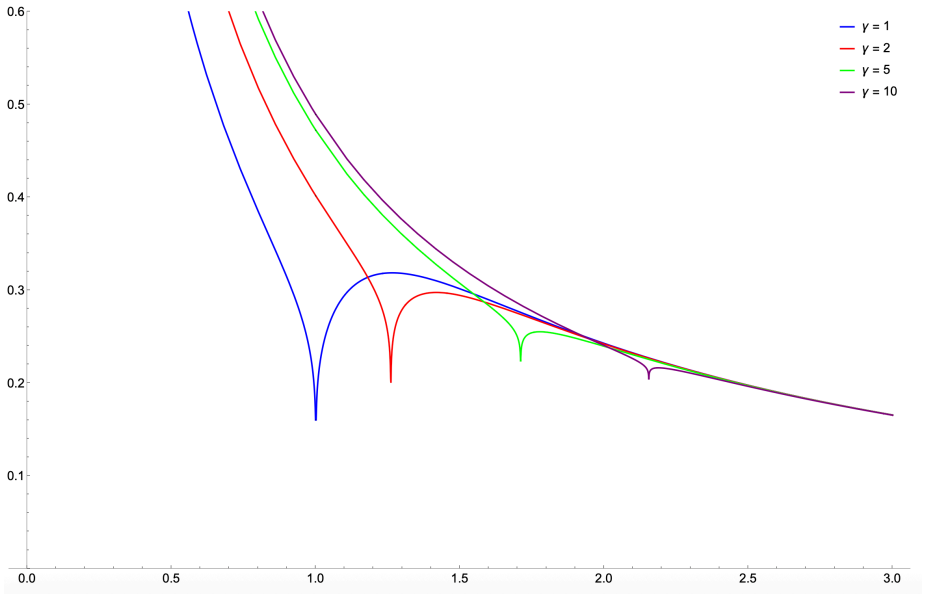


Figure 6: HHF Density of states $g(k)$. The zeros in the HHF density of states $g(k)$ are shown for various values of $\gamma = \frac{R-1}{N-1}$, occuring at the fictitious Fermi wavevectors k_R.

Figure 5 is similar to the graphical representation used by *Blair et al.*[12]. We observe that as we increase γ, the divergent derivative occurs at larger values of k, and therefore higher energies. In this way, we can effectively push the band gap beyond the region in which we are interested (often around the Fermi wavevector) as depicted in Figure 6. This suggests that while a lack of correlation can explain the unphysical band gap in the HF dispersion for jellium, it is not the only method to rectify this issue.

5 Conclusion

We have described the Hartree-Fock and Hyper-Hartree-Fock theories and their methods for obtaining dispersion relations for jellium. We have discussed the generally accepted consensus on a particular shortcoming of the HF method in which an unphysical band gap is predicted at the Fermi wavevector. We show that through the HHF method we can move this band gap arbitrarily far from the Fermi wavevector, effectively diminishing its importance. Our ability to manipulate this band gap implies that it is a mathematical artefact of the structure of the non-local Fock exchange term rather than an intrinsic problem with the method lacking correlation, as is often suggested by the literature.

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